SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO4 – Articulation (BL3)	**Assessment:**	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA	**Total Marks:**	100
CO4 Statement:	*Apply hashing strategies for efficient data storage and sorting algorithms for arrangement of data.*		
Student Name:	K. Kaviya	**Reg. No.:**	192521262
Section / Batch:	C	**Date:**	28/08/2026

Code of Conduct

I K. Kaviya (192521262) *(Name / Reg No)*

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Kaviya

Instructions

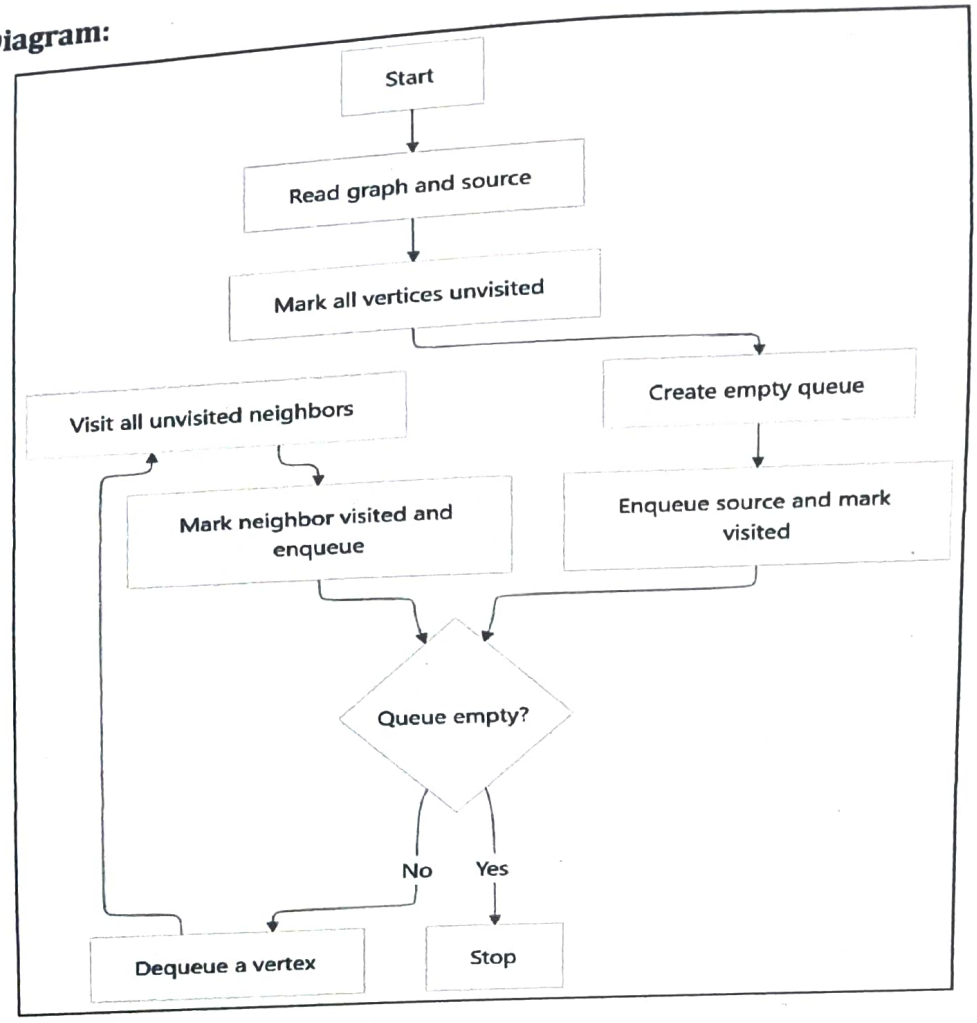
- This test contains 5 Comparative Analysis questions. Total: 100 Marks.
- When a student answer using comparative analysis should be based on four requirements: time complexity, space complexity advantages & limitations, real-world scenarios and operations.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Hashing	Linear Probing & Separate Chaining	1	20
Sorting	Merge Sort & Quick Sort	2	20
Hashing	Quadratic Probing and Double Hashing	3	20
Sorting Algorithms	Complexity Analysis	4	20
Quick Sort	Recursive and Iterative implementations	5	20
	GRAND TOTAL:		100 Marks

Annexure A – Flowchart-To-Code Conversion

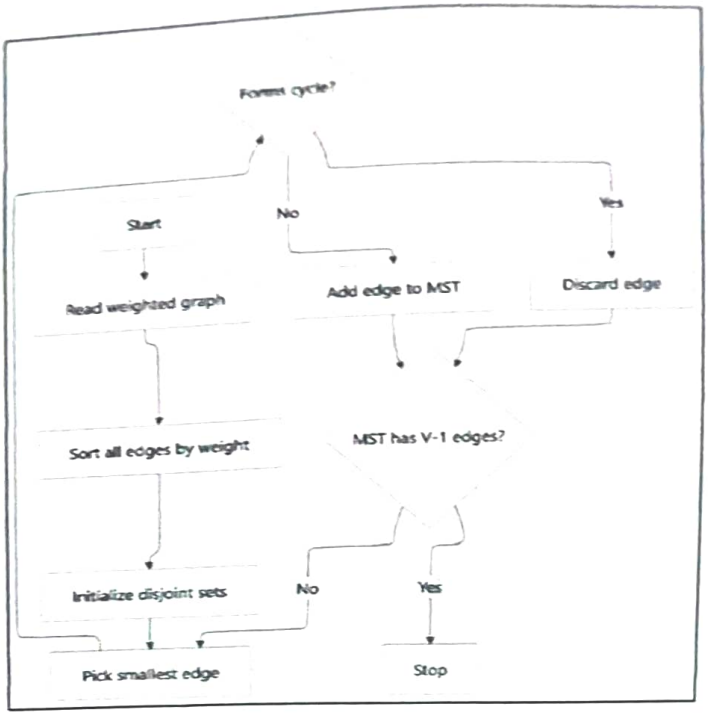
1. Convert the flowchart for Breadth First Search traversal into code. (20 Marks)

Flowchart Diagram:



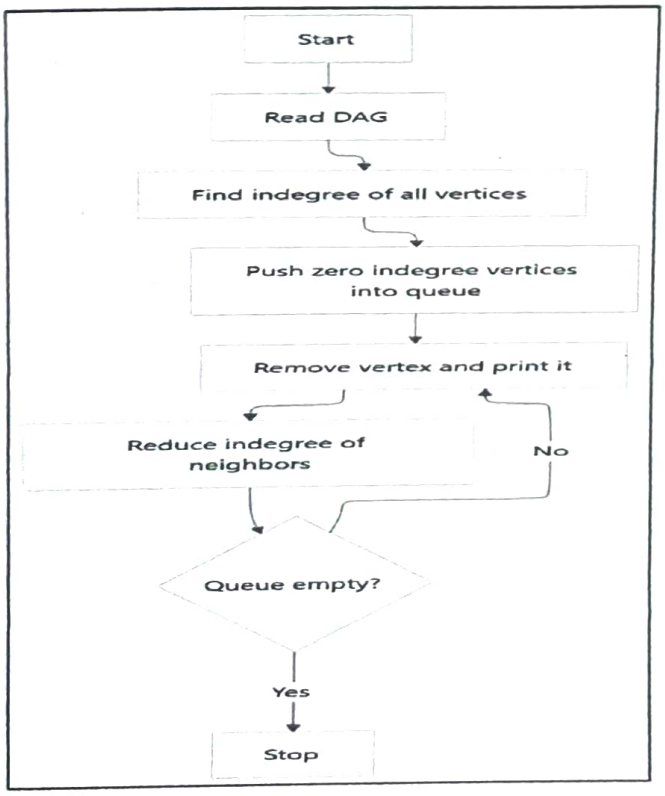
2. Convert the flowchart for Kruskal's algorithm into code. (20 Marks)

Flowchart Diagram:



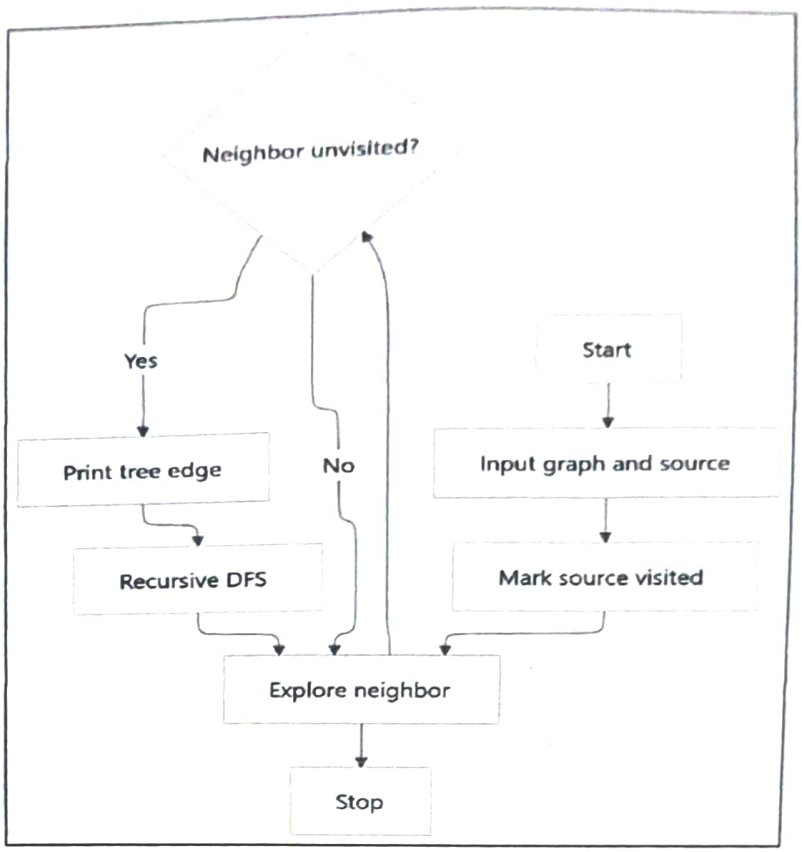
3. Convert the flowchart for topological sort into code. (20 Marks)

Flowchart Diagram:



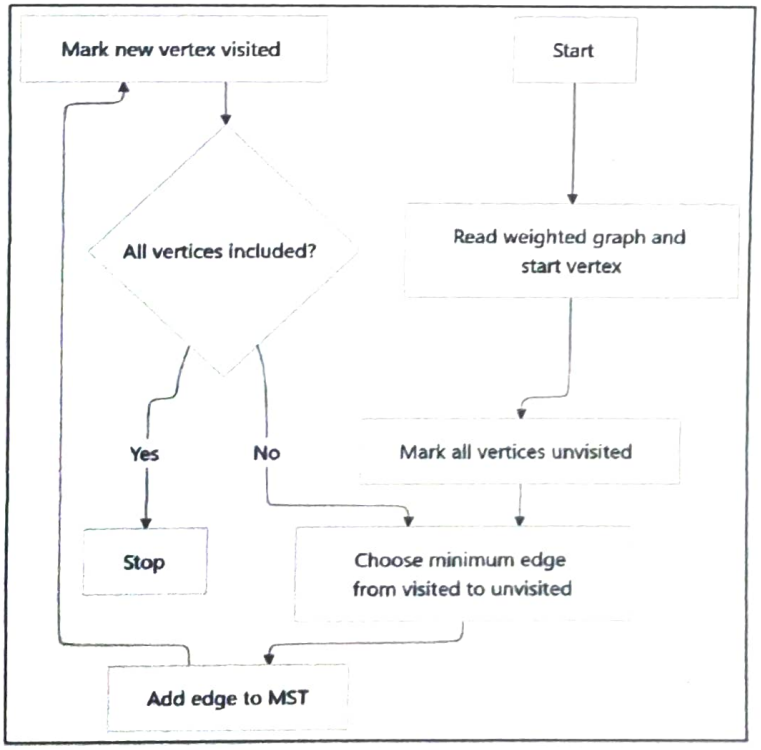
4. Convert the flowchart for generating a DFS tree into code. (20 Marks)

Flowchart Diagram:



5. Convert the flowchart for Prim's algorithm into code. (20 Marks)

Flowchart Diagram:



Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or non-functional code
Code Quality & Readability	20	Clean, modular, well-commented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

05 - Assessment Tool - 02

- Flowchart-to-code-conversion

Name :

Reg No :

1 Breadth First search traversal :

```
def bfs(graph, start):
    visited = [False] * len(graph)
    queue = [start]
    visited[start] = True
    while queue:
        u = queue.pop(0)
        print(u, end = " ")
        for v in graph[u]:
            visited[v] = True
            queue.append(v)
graph = {0: [1,2], 1:[3], 2:[4,5], 3:[], 4:[], 5:[]}
bfs(graph, 0)
bfs(graph, 0)
```

2.

```
def kruskal(n, edges):
    parent = list(range(n))
    def find(x):
        while parent[x] != x:
            x = parent[x]
        return x
```

```
mst = [ ]
for u,v,w in sorted (edgs , key = (lambda x: x[2]):
if find (u) != find (v)
mst . appoad ((u,v,w))
parent [find(u)] = find (v)
return mst
edgs = [(0,1,10) (0,2,6), (0,3,5) (1,3,15) , (2,3,4)]
```

3.

```
def topo- sort (graph, n):
    indeg =[0] * n
for u in graph:
  for v in graph [u]: indeg [v] += 1
q = [ifor i in range(n) if indeg[i]==0]
  while q:
  u=q .pop(0)
print (u, end = ' ')
for v in graph [u];
      in deg [v] = 1
   if indeg [v]==0;  q.append (v)
graph = {0: [1,2] , 1:[3] , 2:[3], 3:[ ]}
topo - sort (graph, u)
```

4. DFS tree:

```
def dfs (graph, u, visited)
    visited [u] = True
    print (u, end = (' '))
    for v in graph [u]:
        if not visited [v]:
            dfs (graph, v, visited)
graph = {0: [1,2], 1:[3], 3:[4,5], 3:[], 4[], 5:[]}
    visited - [False] * 6.
    dfs (graph, 0, visited)
```

5.

```
def prim( graph, start):
    n = len(graph)
visited = [False] * n
visited = [start] = True
        mst = [ ]
for - in range (n-1):
    u = v = 1, w = float ('inf')
        for i in range(n):
        if visited [i]:
    for j, c in graph [i]:
if not visited [j] and c < w
visited [v] = u, v, w = i, j, c
            True.
mst. append (u, v, w), return mst.
```

graph = {

0: [(1,10), (2,6), (3,5)], 1: [(0,10), (3,15)]

2: [(0,6), (3,4)], 3: [(0,5), (1,15), (2,4)]

Undirected